

Personal Profile

Aspiring **Robotics Engineer** combining **robotics** and **AI** to build intelligent, autonomous systems, with skills in **control algorithms**, **sensor fusion**, and **robot navigation** using **ROS** and **Python**. Experienced with **machine learning** and **computer vision** for improving robot perception and decision-making, with **hands-on** experience building mobile robots and robotic manipulators.

Education

Aligarh Muslim University <i>Bachelors of Technology (B.Tech), Electrical Engineering</i>	Aug 2023 – July 2027
Senior Secondary (Class XII), AMU Board — 92.7%	2022
Secondary (Class X), CBSE Board — 87%	2020

Experience

Robotics Engineering Intern	Jun 2026 – Aug 2026
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IISc Bengaluru, Department of Aerospace Engineering

- Engineered a cost-effective, real-time pose estimation system for autonomous drone navigation using a positional camera and a **Raspberry Pi 5**, applying **OpenCV** and **PnP** algorithms to track an asymmetric IR LED array.
- Integrated **ESP32 UWB (Ultra-Wideband)** modules to perform accurate spatial trilateration, establishing precise drone landing zones to complement optical tracking data.
- Developed a hybrid data pipeline utilizing **Kalman filtering** and permutation-based hypothesis testing over UDP telemetry, eliminating coordinate jitter and ensuring stable **6-DoF** tracking during complex flight maneuvers.

Mechatronics Intern

Jan 2026

- Engineered an **IoT-enabled monitoring system** for manual grease injection pumps to enhance leak prevention protocols in oil and gas pipelines.
- Integrated an **ESP32** and **Hall effect sensor** to measure magnetic field variations, enabling precise tracking of lever pump counts and angular displacement.
- Implemented data transmission protocols via **Bluetooth Low Energy (BLE)** to provide real-time kinematics and maintenance logs for remote diagnostics.

Product Development Intern

July 2024 – Oct 2024

- Designed an adaptable orthopedic device for **clubfoot (CTEV)** treatment, prioritizing patient stability and ergonomic comfort.
- Developed patient-specific components using advanced **3D scanning** and additive manufacturing (3D printing) techniques.
- Collaborated with clinicians and engineers to ensure device compliance with medical standards and user-centered design requirements.

Projects

Autonomous Underwater Vehicle (AUV)	Jan 2025
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- Implemented real-time **6-DoF pose estimation** using the **Perspective-n-Point (PnP)** algorithm and fused vision, IMU, and depth data for autonomous navigation and closed-loop trajectory tracking.
- Configured and tuned **Pixhawk** running **ArduSub** firmware for underwater dynamics, optimizing **PID control** across surge, sway, heave, roll, pitch, and yaw.
- Established **Jetson Xavier NX – Pixhawk** communication via **MAVLink** (UART) enabling offboard high-level control on Jetson and low-level stabilization on Pixhawk.
- Integrated and validated an **8-thruster vectored 6-DoF configuration**, including ESC calibration, motor mixing, failsafe tuning, power distribution, and sensor interfacing (I2C/UART).

Autonomous Surface Vehicle (ASV)

Nov 2024

- Designed the end-to-end electronics architecture for a differential-drive (dual-thruster) ASV, integrating a Jetson Xavier NX, IMUs, and an **RTK GPS** module to achieve high-precision localization.
- Developed a custom closed-loop control layer utilizing a carrot-following algorithm and **PID tuning**, processing real-time GPS position feedback to ensure smooth and highly accurate autonomous waypoint navigation.
- Engineered a dual-layer communication framework utilizing **TCP/UDP** for real-time telemetry visualization at

the ground station, alongside a long-range **LoRa** radio link for RTK GPS corrections and manual overrides.

Remotely Operated Underwater Vehicle (ROV)

Nov 2024

- Developed joystick-based teleoperation using **Raspberry Pi** (PWM-based ESC control) with **MAVLink** telemetry integration.
- Configured **Pixhawk + ArduSub** for manual and assisted modes (depth-hold, attitude stabilization) in an **8-thruster vectored 6-DoF** frame.
- Ensured reliable communication between Raspberry Pi, Pixhawk, ESCs, IMU, camera, and onboard sensors; performed **PID tuning** and motor mixing validation for stable underwater operation.

Personalized Electronic Door Lock

Dec 2023

- Designed and developed a **CAD model of a custom electronic locking mechanism** for personal use.
- Utilized an **ESP32**, **stepper motor**, and motor driver to enable **wireless control** of the door lock.
- Implemented a user-friendly solution allowing the door to be locked and unlocked remotely.

Technical Skills

Programming & Tools: Python, Git/GitHub, Linux (Ubuntu), Docker

Robotics & Simulation: ROS 2 (Humble/Jazzy), Gazebo, Nav2, Rviz2, URDF, OpenCV, TensorFlow

Control & Navigation: 6-DoF Control, State/Pose Estimation (PnP), Sensor Fusion (EKF), PID Tuning, Path Planning

Embedded Systems: Pixhawk (ArduSub), MAVLink, 8051 Microcontroller, NVIDIA Jetson, Raspberry Pi, UART/I2C/SPI, PWM/ESC

Design & Manufacturing: Fusion 360 (CAD), Cura (3D Printing)

Achievements & Awards

Shortlisted Team – SAUVC, Singapore (Jan 2025) — Selected as 1 of 111 global teams for the Singapore Autonomous Underwater Vehicle Challenge.

Best Innovation Award – MTS Tech Symposium, IIT Madras (March 2025) — Recognized for innovative work on the team's Autonomous Underwater Vehicle (AUV).

2nd Place – AMUROVC 3.0, AMU (Nov 2024) — Secured 2nd position for the team's Remotely Operated Underwater Vehicle (ROV).

IEEE DELCON 2025 – Jamia Millia Islamia (Oct 2025) — Presented AUV project at the IEEE Delhi Section Conference.

TechnoXian 2025 – RoboHockey Competition (Aug 2025) — Built a robot for RoboHockey; qualified in the first round among national and international teams.

RoboSoccer Tryst, IIT Delhi (March 2025) — Designed and built a remote-controlled soccer-playing robot.

Leadership

Electronics Team Lead, AUV-ZHCET

Aug 2024 – July 2025

- Contributed to the **electronics design** and **control systems** of both ROV and AUV platforms.
- Developed and implemented the **pose estimation** algorithm for the AUV to improve underwater navigation and localization.
- Programmed core **control algorithms** and integrated the **AUV's electronic architecture**, including **sensor interfacing** and actuator control for stable underwater maneuvering.
- Delivered a **four-day practical workshop** on designing and building **robotic arms**, training **120+ participants** in mechanical design, control, and programming.
- Delivered a **one-day practical** workshop on designing and building RC cars, training **170+ participants** in applied robotics and control systems.

Volunteer, AMU Roboclub

Aug 2024 – July 2025

- Organized and conducted a hands-on **workshop on Arduino** for beginners and robotics enthusiasts.
- Contributed to ongoing projects, including **RoboSoccer**, by providing technical input and design suggestions.
- Coordinated **VERCERA**, the club's flagship robotics event, managing logistics, outreach, and event execution.